

UNIVERSIDAD NACIONAL DE COLOMBIA

Facultad de Minas

Departamento de Ciencias de la Computación y de la Decisión

Ingeniería de Sistemas e Informática

Programación Orientada a Objetos (2026-2S)

Actividad 3: Interfaces Gráficas (GUI)

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Fecha de Entrega:
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1 Repositorio de GitHub

El código fuente compilable en Java y la estructura completa del proyecto se encuentran alojados en el siguiente repositorio público de GitHub:

https://github.com/alejandrogb09/Activity_3_POO_2026-1

Nota: El archivo ***`Informacion.txt`*** *del repositorio contiene la información institucional requerida (nombre completo de la universidad, nombre de la actividad, nombre completo del estudiante y nombre completo del docente).*

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2 Ejercicio 8.2: Cálculo de Promedio y Desviación Estándar de Notas

2.1 Enunciado

Desarrollar un programa gráfico en Swing que reciba 5 notas de un estudiante y calcule su promedio, desviación estándar, nota mayor y nota menor. El sistema debe validar que los campos no estén vacíos ni contengan valores fuera del rango permitido $(0.0 - 5.0)$.

2.2 Diagrama de Casos de Uso UML

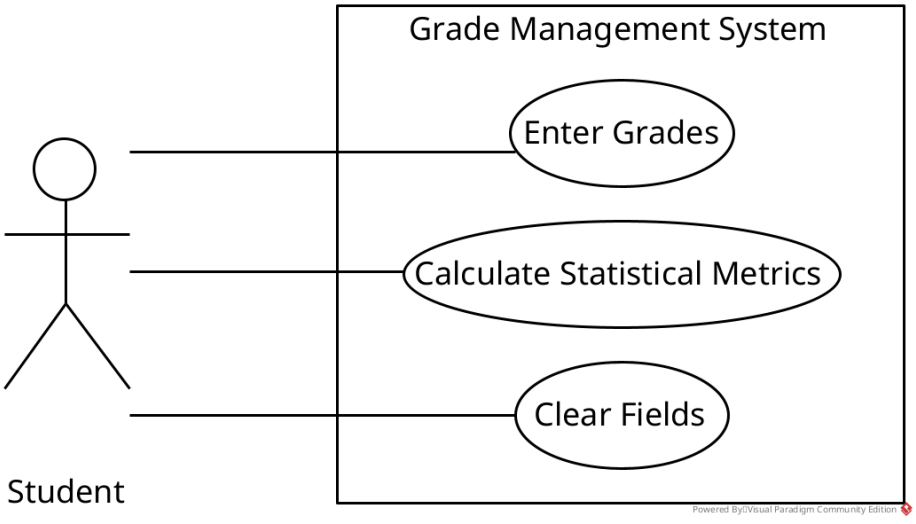


Figure 1: Diagrama de Casos de Uso - Ejercicio 8.2

2.3 Diagrama de Clases UML

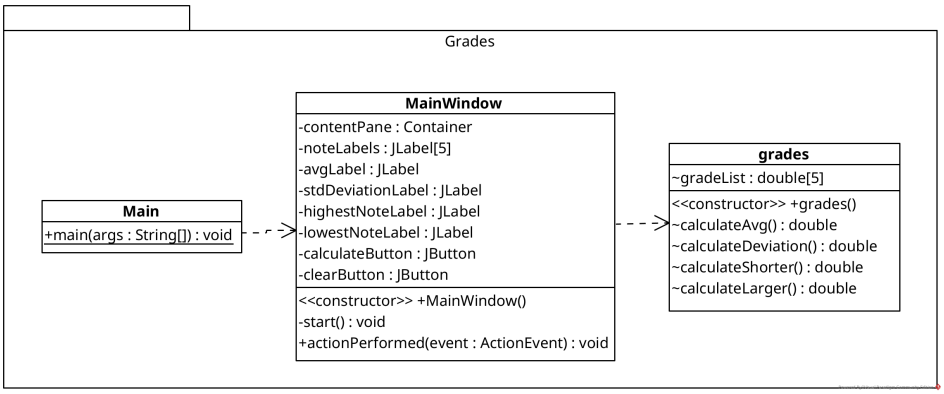


Figure 2: Diagrama de Clases UML - Ejercicio 8.2

2.4 Diagrama de Objetos UML

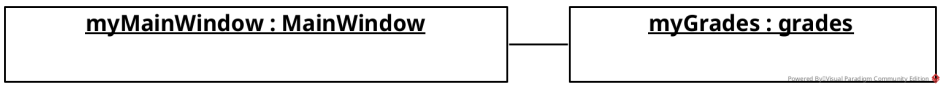


Figure 3: Diagrama de Objetos UML - Ejercicio 8.2

2.5 Interfaz Gráfica de Usuario (GUI)

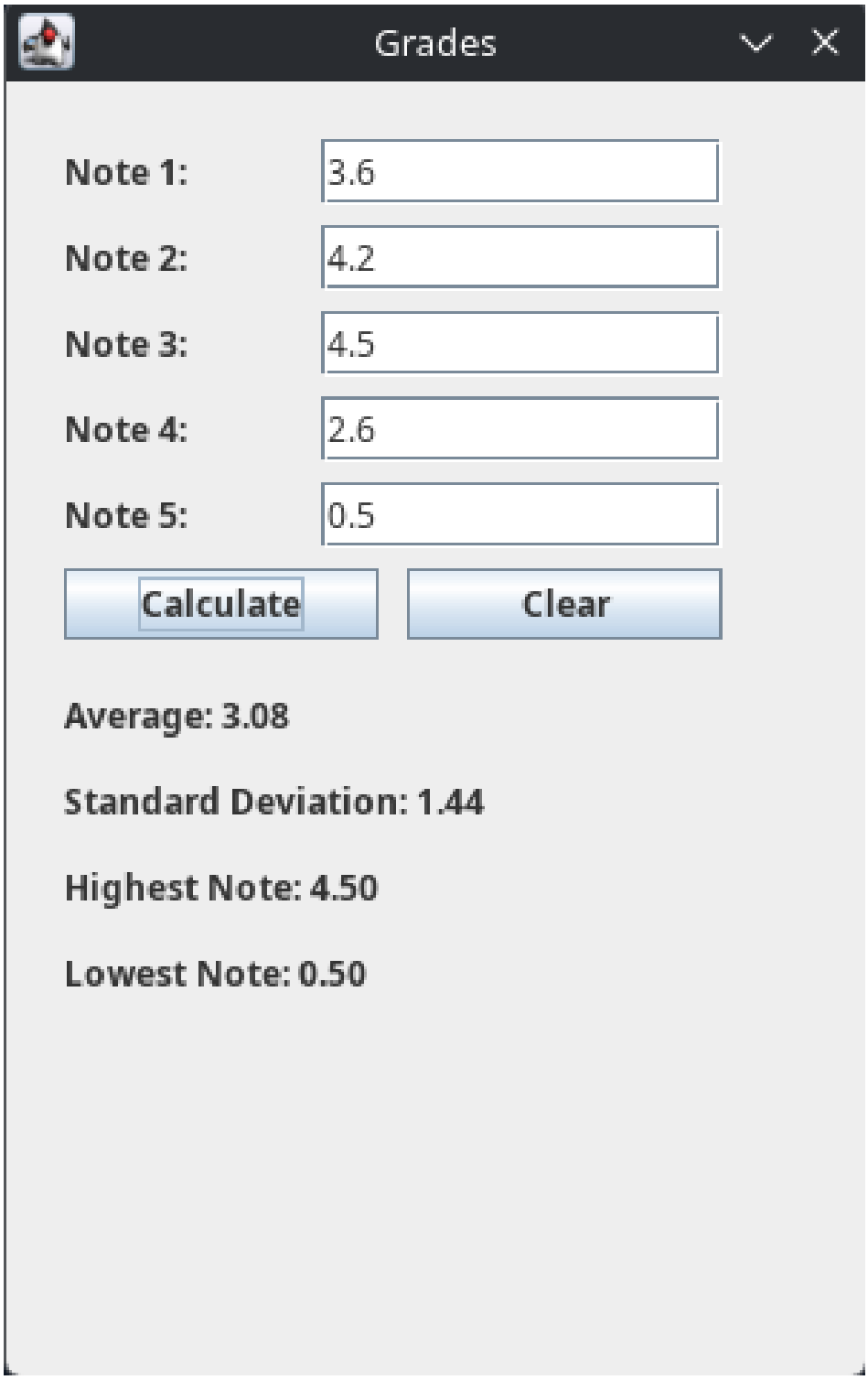


Figure 4: Interfaz Gráfica de Usuario - Ejercicio 8.2

2.6 Código Fuente en Java

2.6.1 Clase de Lógica / Modelo (`Grades.java`)

```java
package Grades;

public class Grades {
    double[] gradeList;

    public Grades() {
        gradeList = new double[5];
    }

    double calculateAvg() {
        double sum = 0;
        for (int i=0; i < gradeList.length; i++) {
            sum = sum + gradeList[i];
        }
        return (sum / gradeList.length);
    }

    double calculateDeviation() {
        double avg = calculateAvg();
        double sum = 0;
        for(int i=0; i < gradeList.length; i++) {
            sum += Math.pow(gradeList[i] - avg, 2);
        }
        return Math.sqrt(sum/gradeList.length);
    }

    double calculateShorter() {
        double shorter = gradeList[0];
        for (int i=0; i < gradeList.length; i++) {
            if(gradeList[i] < shorter) {
                shorter = gradeList[i];
            }
        }
        return shorter;
    }

    double calculateLarger() {
        double larger = gradeList[0];
        for (int i=0; i < gradeList.length; i++) {
            if (gradeList[i] > larger) {
                larger = gradeList[i];
            }
        }
        return larger;
    }
}
```

Listing 1: Clase de Modelo para el cálculo de Notas

2.6.2 Clase de Vista / Ventana Principal (`MainWindow.java`)

```
package Grades;
import java.awt.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;
import javax.swing.*;

public class MainWindow extends JFrame implements ActionListener {
    private Container contentPane;
    private JLabel[] noteLabels;
    private JTextField[] noteFields;
    private JLabel avgLabel, stdDeviationLabel, highestNoteLabel,
lowestNoteLabel;
    private JButton calculateButton, clearButton;

    public MainWindow() {
        start();
        setTitle("Grades");
        setSize(300, 480);
        setLocationRelativeTo(null);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setResizable(false);
    }

    private void start() {
        contentPane = getContentPane();
        contentPane.setLayout(null);

        noteLabels = new JLabel[5];
        noteFields = new JTextField[5];

        int yPosition = 20;
        for (int i = 0; i < 5; i++) {
            noteLabels[i] = new JLabel("Note " + (i + 1) + ":");
            noteLabels[i].setBounds(20, yPosition, 80, 23);
            contentPane.add(noteLabels[i]);

            noteFields[i] = new JTextField();
            noteFields[i].setBounds(110, yPosition, 140, 23);
            contentPane.add(noteFields[i]);

            yPosition += 30;
        }

        calculateButton = new JButton("Calculate");
        calculateButton.setBounds(20, yPosition, 110, 25);
        calculateButton.addActionListener(this);
        contentPane.add(calculateButton);

        clearButton = new JButton("Clear");
        clearButton.setBounds(140, yPosition, 110, 25);
        clearButton.addActionListener(this);
        contentPane.add(clearButton);
```

```
52
53                  avgLabel = new JLabel("Average: ");
54                  avgLabel.setBounds(20, yPosition + 40, 250, 23);
55                  contentPane.add(avgLabel);
56
57                  stdDeviationLabel = new JLabel("Standard Deviation: ");
58                  stdDeviationLabel.setBounds(20, yPosition + 70, 250, 23);
59                  contentPane.add(stdDeviationLabel);
60
61                  highestNoteLabel = new JLabel("Highest Note: ");
62                  highestNoteLabel.setBounds(20, yPosition + 100, 250, 23);
63                  contentPane.add(highestNoteLabel);
64
65                  lowestNoteLabel = new JLabel("Lowest Note: ");
66                  lowestNoteLabel.setBounds(20, yPosition + 130, 250, 23);
67                  contentPane.add(lowestNoteLabel);
68              }
69
70              @Override
71              public void actionPerformed(ActionEvent event) {
72                  if (event.getSource() == calculateButton) {
73                      Grades grades = new Grades();
74                      for (int i = 0; i < 5; i++) {
75                          String rawInput = noteFields[i].getText().trim();
76
77                          if (rawInput.isEmpty()) {
78                              JOptionPane.showMessageDialog(this,
79                              "Note " + (i + 1) + " has not been entered.
    All 5 notes are mandatory.",
80                              "Missing Data",
81                              JOptionPane.WARNING_MESSAGE);
82                              noteFields[i].requestFocus();
83                              return;
84                          }
85
86                          try {
87                              grades.gradeList[i] = Double.parseDouble(
    rawInput);
88                          } catch (NumberFormatException e) {
89                              JOptionPane.showMessageDialog(this,
90                              "The input '" + rawInput + "' in Note " + (i +
     1) + " is not numeric.\nPlease enter a valid number.",
91                              "Invalid Format",
92                              JOptionPane.ERROR_MESSAGE);
93                              noteFields[i].requestFocus();
94                              return;
95                          }
96                      }
97
98                      avgLabel.setText("Average: " + String.format("%.2f",
    grades.calculateAvg()));
99                      stdDeviationLabel.setText("Standard Deviation: " +
    String.format("%.2f", grades.calculateDeviation()));
```

```
                    highestNoteLabel.setText("Highest Note: " + String.
    format("%.2f", grades.calculateLarger()));
                    lowestNoteLabel.setText("Lowest Note: " + String.
    format("%.2f", grades.calculateShorter()));
                }

                if (event.getSource() == clearButton) {
                    for (int i = 0; i < 5; i++) {
                        noteFields[i].setText("");
                    }
                    avgLabel.setText("Average: ");
                    stdDeviationLabel.setText("Standard Deviation: ");
                    highestNoteLabel.setText("Highest Note: ");
                    lowestNoteLabel.setText("Lowest Note: ");
                }
            }
        }

```

Listing 2: Clase de Interfaz Gráfica en Swing

2.6.3 Clase Principal (`Main.java`)

```
        package Grades;

        public class Main {
            public static void main(String[] args) {
                MainWindow myMainWindow;
                myMainWindow = new MainWindow();
                myMainWindow.setVisible(true);
            }

        }

```

Listing 3: Clase principal desde dónde se ejecuta el programa.

3 Ejercicio 8.3: Cálculo de Volumen y Superficie de Figuras Geométricas

3.1 Enunciado

Diseñar una aplicación con interfaz gráfica Swing que permita seleccionar entre tres figuras geométricas tridimensionales (Cilindro, Esfera y Pirámide), desplegar sus respectivas ventanas de ingreso de dimensiones y calcular su volumen y superficie.

3.2 Diagrama de Casos de Uso UML

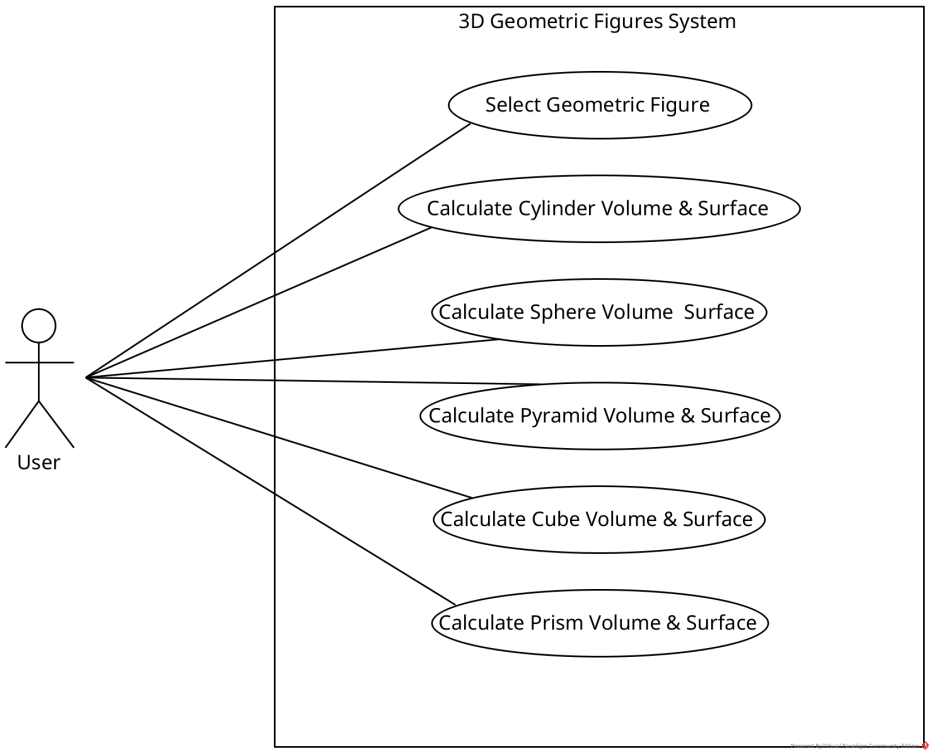


Figure 5: Diagrama de Casos de Uso - Ejercicio 8.3

3.3 Diagrama de Clases UML

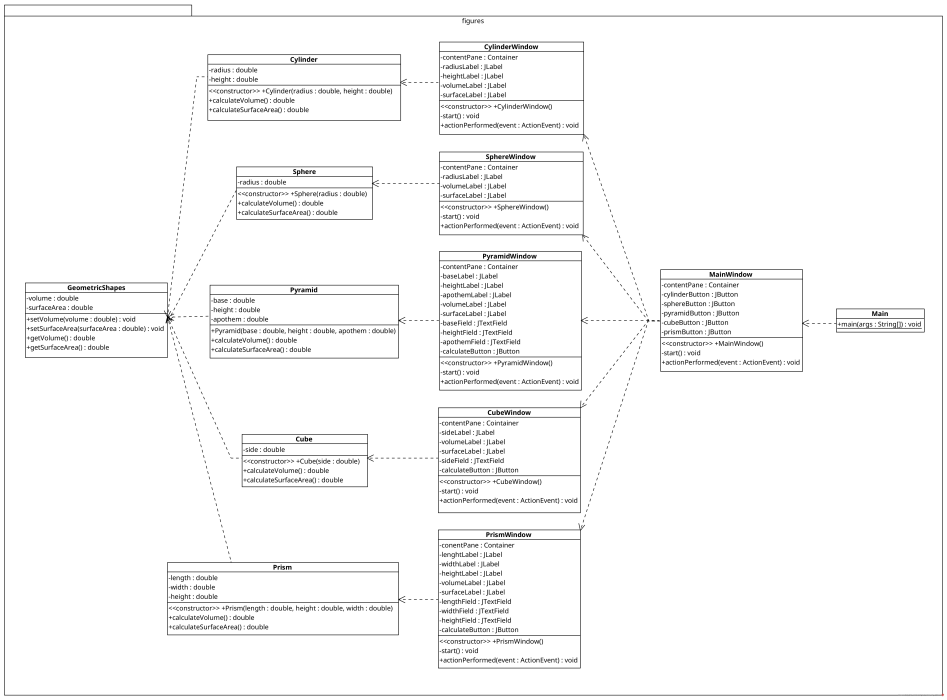


Figure 6: Diagrama de Clases UML - Ejercicio 8.3

3.4 Diagrama de Objetos UML

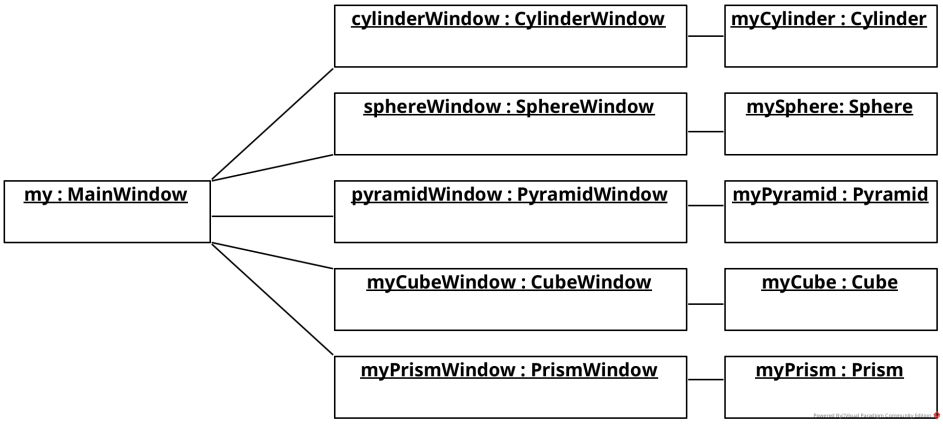


Figure 7: Diagrama de Objetos UML - Ejercicio 8.3

3.5 Interfaz Gráfica de Usuario (GUI)

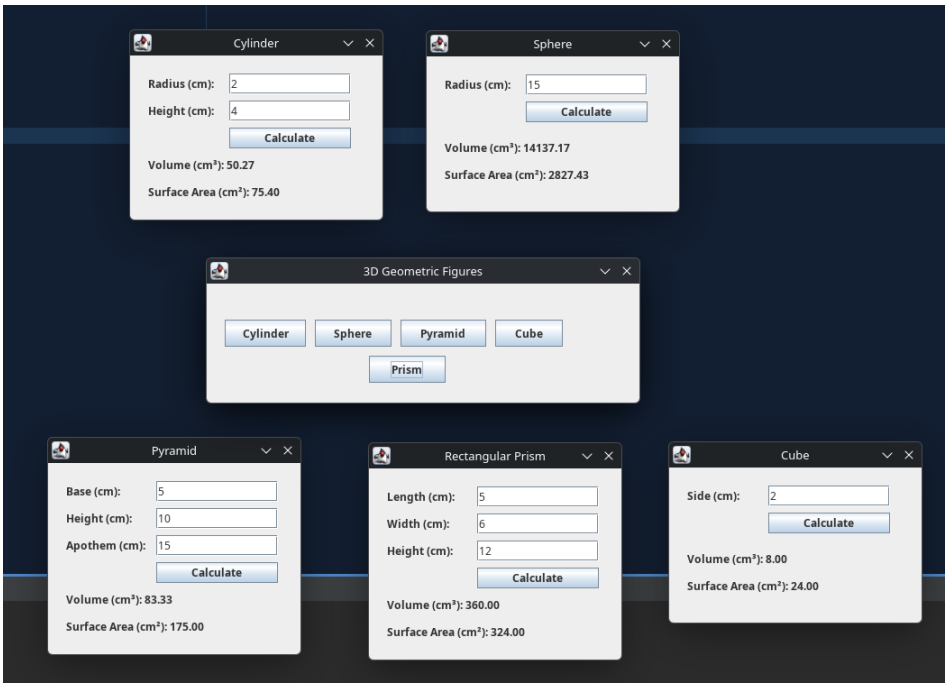


Figure 8: Interfaz Gráfica de Usuario - Ejercicio 8.3

3.6 Código Fuente en Java

3.6.1 Clases de Modelo (`Geometric_Shapes.java`, `Cylinder.java`, etc.)

```
package figures;

public class Geometric_Shapes {
    private double volume;
    private double surfaceArea;

    public void setVolume(double volume) {
        this.volume = volume;
    }

    public void setSurfaceArea(double surfaceArea) {
        this.surfaceArea = surfaceArea;
    }

    public double getVolume() {
        return this.volume;
    }

    public double getSurfaceArea() {
        return this.surfaceArea;
    }
}
```

```
23
24          public class Cylinder extends Geometric_Shapes {
25              private double radius;
26              private double height;
27
28              public Cylinder(double radius, double height) {
29                  this.radius = radius;
30                  this.height = height;
31                  this.setVolume(calculateVolume());
32                  this.setSurfaceArea(calculateSurfaceArea());
33              }
34
35              public double calculateVolume() {
36                  return Math.PI * height * Math.pow(radius, 2.0);
37              }
38
39              public double calculateSurfaceArea() {
40                  double sideAreaA = 2.0 * Math.PI * radius * height;
41                  double sideAreaB = 2.0 * Math.PI * Math.pow(radius, 2.0);
42                  return sideAreaA + sideAreaB;
43              }
44          }
45
46          public class Sphere extends Geometric_Shapes {
47              private double radius;
48
49              public Sphere(double radius) {
50                  this.radius = radius;
51                  this.setVolume(calculateVolume());
52                  this.setSurfaceArea(calculateSurfaceArea());
53              }
54
55              public double calculateVolume() {
56                  return (4.0 / 3.0) * Math.PI * Math.pow(this.radius, 3.0);
57              }
58
59              public double calculateSurfaceArea() {
60                  return 4.0 * Math.PI * Math.pow(this.radius, 2.0);
61              }
62          }
63
64          public class Pyramid extends Geometric_Shapes {
65              private double base;
66              private double height;
67              private double apothem;
68
69              public Pyramid(double base, double height, double apothem) {
70                  this.base = base;
71                  this.height = height;
72                  this.apothem = apothem;
73                  this.setVolume(calculateVolume());
74                  this.setSurfaceArea(calculateSurfaceArea());
75              }
76
```

```
77              public double calculateVolume() {
78                  return (Math.pow(base, 2.0) * height) / 3.0;
79              }
80
81              public double calculateSurfaceArea() {
82                  double baseArea = Math.pow(base, 2.0);
83                  double sideArea = 2.0 * base * apothem;
84                  return baseArea + sideArea;
85              }
86          }
87
88          package figures;
89
90          public class Cube extends Geometric_Shapes {
91              private double side;
92
93              public Cube(double side) {
94                  this.side = side;
95                  this.setVolume(calculateVolume());
96                  this.setSurfaceArea(calculateSurfaceArea());
97              }
98
99              public double calculateVolume() {
100                 return Math.pow(side, 3.0);
101             }
102
103             public double calculateSurfaceArea() {
104                 return 6.0 * Math.pow(side, 2.0);
105             }
106         }
107
108         public class Prism extends Geometric_Shapes {
109             private double length;
110             private double width;
111             private double height;
112
113             public Prism(double length, double width, double height) {
114                 this.length = length;
115                 this.width = width;
116                 this.height = height;
117                 this.setVolume(calculateVolume());
118                 this.setSurfaceArea(calculateSurfaceArea());
119             }
120
121             public double calculateVolume() {
122                 return length * width * height;
123             }
124
125             public double calculateSurfaceArea() {
126                 return 2.0 * (length * width + length * height + width *
    height);
127             }
128         }
129
```

```
130
```

Listing 4: Jerarquía de Clases de Modelo

3.6.2 Clase de Menú y Ventanas Swing (`MainWindow.java`)

```
1         package figures;
2
3         import javax.swing.*;
4         import java.awt.*;
5         import java.awt.event.ActionEvent;
6         import java.awt.event.ActionListener;
7
8         public class Main {
9             public static void main(String[] args) {
10                MainWindow myMainWindow = new MainWindow();
11                myMainWindow.setVisible(true);
12            }
13        }
14
15        public class MainWindow extends JFrame implements ActionListener {
16            private Container contentPane;
17            private JButton cylinderButton, sphereButton, pyramidButton,
    cubeButton, prismButton;
18
19            public MainWindow() {
20                start();
21                setTitle("3D Geometric Figures");
22                setSize(480, 160);
23                setLocationRelativeTo(null);
24                setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
25                setResizable(false);
26            }
27
28            private void start() {
29                contentPane = getContentPane();
30                contentPane.setLayout(null);
31
32                cylinderButton = new JButton("Cylinder");
33                cylinderButton.setBounds(20, 40, 90, 30);
34                cylinderButton.addActionListener(this);
35
36                sphereButton = new JButton("Sphere");
37                sphereButton.setBounds(120, 40, 85, 30);
38                sphereButton.addActionListener(this);
39
40                pyramidButton = new JButton("Pyramid");
41                pyramidButton.setBounds(215, 40, 95, 30);
42                pyramidButton.addActionListener(this);
43
44                cubeButton = new JButton("Cube");
45                cubeButton.setBounds(320, 40, 75, 30);
46                cubeButton.addActionListener(this);
```

```
47
48              prismButton = new JButton("Prism");
49              prismButton.setBounds(180, 80, 85, 30);
50              prismButton.addActionListener(this);
51
52              contentPane.add(cylinderButton);
53              contentPane.add(sphereButton);
54              contentPane.add(pyramidButton);
55              contentPane.add(cubeButton);
56              contentPane.add(prismButton);
57          }
58
59          @Override
60          public void actionPerformed(ActionEvent event) {
61              if (event.getSource() == cylinderButton) {
62                  CylinderWindow cylinderWindow = new CylinderWindow();
63                  cylinderWindow.setVisible(true);
64              }
65              if (event.getSource() == sphereButton) {
66                  SphereWindow sphereWindow = new SphereWindow();
67                  sphereWindow.setVisible(true);
68              }
69              if (event.getSource() == pyramidButton) {
70                  PyramidWindow pyramidWindow = new PyramidWindow();
71                  pyramidWindow.setVisible(true);
72              }
73              if (event.getSource() == cubeButton) {
74                  CubeWindow cubeWindow = new CubeWindow();
75                  cubeWindow.setVisible(true);
76              }
77              if (event.getSource() == prismButton) {
78                  PrismWindow prismWindow = new PrismWindow();
79                  prismWindow.setVisible(true);
80              }
81          }
82      }
83
84      public class CylinderWindow extends JFrame implements
    ActionListener {
85          private Container contentPane;
86          private JLabel radiusLabel, heightLabel, volumeLabel,
    surfaceLabel;
87          private JTextField radiusField, heightField;
88          private JButton calculateButton;
89
90          public CylinderWindow() {
91              start();
92              setTitle("Cylinder");
93              setSize(280, 210);
94              setLocationRelativeTo(null);
95              setResizable(false);
96          }
97
98          private void start() {
```

```
                contentPane = getContentPane();
                contentPane.setLayout(null);

                radiusLabel = new JLabel("Radius (cm):");
                radiusLabel.setBounds(20, 20, 135, 23);
                radiusField = new JTextField();
                radiusField.setBounds(110, 20, 135, 23);

                heightLabel = new JLabel("Height (cm):");
                heightLabel.setBounds(20, 50, 135, 23);
                heightField = new JTextField();
                heightField.setBounds(110, 50, 135, 23);

                calculateButton = new JButton("Calculate");
                calculateButton.setBounds(110, 80, 135, 23);
                calculateButton.addActionListener(this);

                volumeLabel = new JLabel("Volume (cm³):");
                volumeLabel.setBounds(20, 110, 220, 23);

                surfaceLabel = new JLabel("Surface Area (cm²):");
                surfaceLabel.setBounds(20, 140, 220, 23);

                contentPane.add(radiusLabel);
                contentPane.add(radiusField);
                contentPane.add(heightLabel);
                contentPane.add(heightField);
                contentPane.add(calculateButton);
                contentPane.add(volumeLabel);
                contentPane.add(surfaceLabel);
            }

            @Override
            public void actionPerformed(ActionEvent event) {
                if (event.getSource() == calculateButton) {
                    try {
                        double r = Double.parseDouble(radiusField.getText
    ().trim());
                        double h = Double.parseDouble(heightField.getText
    ().trim());
                        Cylinder cylinder = new Cylinder(r, h);
                        volumeLabel.setText("Volume (cm³): " + String.
    format("%.2f", cylinder.getVolume()));
                        surfaceLabel.setText("Surface Area (cm²): " +
    String.format("%.2f", cylinder.getSurfaceArea()));
                    } catch (Exception e) {
                        JOptionPane.showMessageDialog(this, "Null field or
     invalid number format", "Error", JOptionPane.ERROR_MESSAGE);
                    }
                }
            }
        }

```

```
147         public class SphereWindow extends JFrame implements ActionListener
    {
148             private Container contentPane;
149             private JLabel radiusLabel, volumeLabel, surfaceLabel;
150             private JTextField radiusField;
151             private JButton calculateButton;
152 
153             public SphereWindow() {
154                 start();
155                 setTitle("Sphere");
156                 setSize(280, 200);
157                 setLocationRelativeTo(null);
158                 setResizable(false);
159             }
160 
161             private void start() {
162                 contentPane = getContentPane();
163                 contentPane.setLayout(null);
164 
165                 radiusLabel = new JLabel("Radius (cm):");
166                 radiusLabel.setBounds(20, 20, 135, 23);
167                 radiusField = new JTextField();
168                 radiusField.setBounds(110, 20, 135, 23);
169 
170                 calculateButton = new JButton("Calculate");
171                 calculateButton.setBounds(110, 50, 135, 23);
172                 calculateButton.addActionListener(this);
173 
174                 volumeLabel = new JLabel("Volume (cm^3):");
175                 volumeLabel.setBounds(20, 90, 220, 23);
176 
177                 surfaceLabel = new JLabel("Surface Area (cm^2):");
178                 surfaceLabel.setBounds(20, 120, 220, 23);
179 
180                 contentPane.add(radiusLabel);
181                 contentPane.add(radiusField);
182                 contentPane.add(calculateButton);
183                 contentPane.add(volumeLabel);
184                 contentPane.add(surfaceLabel);
185             }
186 
187             @Override
188             public void actionPerformed(ActionEvent event) {
189                 if (event.getSource() == calculateButton) {
190                     try {
191                         double r = Double.parseDouble(radiusField.getText
    ().trim());
192                         Sphere sphere = new Sphere(r);
193                         volumeLabel.setText("Volume (cm^3): " + String.
    format("%.2f", sphere.getVolume()));
194                         surfaceLabel.setText("Surface Area (cm^2): " +
    String.format("%.2f", sphere.getSurfaceArea()));
195                     } catch (Exception e) {
```

```
196                         JOptionPane.showMessageDialog(this, "Null field or
    invalid number format", "Error", JOptionPane.ERROR_MESSAGE);
197                     }
198                 }
199             }
200         }
201 
202         public class PyramidWindow extends JFrame implements
    ActionListener {
203             private Container contentPane;
204             private JLabel baseLabel, heightLabel, apothemLabel,
    volumeLabel, surfaceLabel;
205             private JTextField baseField, heightField, apothemField;
206             private JButton calculateButton;
207 
208             public PyramidWindow() {
209                 start();
210                 setTitle("Pyramid");
211                 setSize(280, 240);
212                 setLocationRelativeTo(null);
213                 setResizable(false);
214             }
215 
216             private void start() {
217                 contentPane = getContentPane();
218                 contentPane.setLayout(null);
219 
220                 baseLabel = new JLabel("Base (cm):");
221                 baseLabel.setBounds(20, 20, 135, 23);
222                 baseField = new JTextField();
223                 baseField.setBounds(120, 20, 135, 23);
224 
225                 heightLabel = new JLabel("Height (cm):");
226                 heightLabel.setBounds(20, 50, 135, 23);
227                 heightField = new JTextField();
228                 heightField.setBounds(120, 50, 135, 23);
229 
230                 apothemLabel = new JLabel("Apothem (cm):");
231                 apothemLabel.setBounds(20, 80, 135, 23);
232                 apothemField = new JTextField();
233                 apothemField.setBounds(120, 80, 135, 23);
234 
235                 calculateButton = new JButton("Calculate");
236                 calculateButton.setBounds(120, 110, 135, 23);
237                 calculateButton.addActionListener(this);
238 
239                 volumeLabel = new JLabel("Volume (cm^3):");
240                 volumeLabel.setBounds(20, 140, 220, 23);
241 
242                 surfaceLabel = new JLabel("Surface Area (cm^2):");
243                 surfaceLabel.setBounds(20, 170, 220, 23);
244 
245                 contentPane.add(baseLabel);
246                 contentPane.add(baseField);
```

```
247                 contentPane.add(heightLabel);
248                 contentPane.add(heightField);
249                 contentPane.add(apothemLabel);
250                 contentPane.add(apothemField);
251                 contentPane.add(calculateButton);
252                 contentPane.add(volumeLabel);
253                 contentPane.add(surfaceLabel);
254             }
255 
256             @Override
257             public void actionPerformed(ActionEvent event) {
258                 if (event.getSource() == calculateButton) {
259                     try {
260                         double b = Double.parseDouble(baseField.getText().
    trim());
261                         double h = Double.parseDouble(heightField.getText
    ().trim());
262                         double a = Double.parseDouble(apothemField.getText
    ().trim());
263                         Pyramid pyramid = new Pyramid(b, h, a);
264                         volumeLabel.setText("Volume (cm^3): " + String.
    format("%.2f", pyramid.getVolume()));
265                         surfaceLabel.setText("Surface Area (cm^2): " +
    String.format("%.2f", pyramid.getSurfaceArea()));
266                     } catch (Exception e) {
267                         JOptionPane.showMessageDialog(this, "Null field or
     invalid number format", "Error", JOptionPane.ERROR_MESSAGE);
268                     }
269                 }
270             }
271         }
272 
273         public class CubeWindow extends JFrame implements ActionListener {
274             private Container contentPane;
275             private JLabel sideLabel, volumeLabel, surfaceLabel;
276             private JTextField sideField;
277             private JButton calculateButton;
278 
279             public CubeWindow() {
280                 start();
281                 setTitle("Cube");
282                 setSize(280, 200);
283                 setLocationRelativeTo(null);
284                 setResizable(false);
285             }
286 
287             private void start() {
288                 contentPane = getContentPane();
289                 contentPane.setLayout(null);
290 
291                 sideLabel = new JLabel("Side (cm):");
292                 sideLabel.setBounds(20, 20, 135, 23);
293                 sideField = new JTextField();
294                 sideField.setBounds(110, 20, 135, 23);
```

```
295
296                 calculateButton = new JButton("Calculate");
297                 calculateButton.setBounds(110, 50, 135, 23);
298                 calculateButton.addActionListener(this);
299
300                 volumeLabel = new JLabel("Volume (cm³):");
301                 volumeLabel.setBounds(20, 90, 220, 23);
302
303                 surfaceLabel = new JLabel("Surface Area (cm²):");
304                 surfaceLabel.setBounds(20, 120, 220, 23);
305
306                 contentPane.add(sideLabel);
307                 contentPane.add(sideField);
308                 contentPane.add(calculateButton);
309                 contentPane.add(volumeLabel);
310                 contentPane.add(surfaceLabel);
311             }
312
313             @Override
314             public void actionPerformed(ActionEvent event) {
315                 if (event.getSource() == calculateButton) {
316                     try {
317                         double l = Double.parseDouble(sideField.getText().
    trim());
318                         Cube cube = new Cube(l);
319                         volumeLabel.setText("Volume (cm³): " + String.
    format("%.2f", cube.getVolume()));
320                         surfaceLabel.setText("Surface Area (cm²): " +
    String.format("%.2f", cube.getSurfaceArea()));
321                     } catch (Exception e) {
322                         JOptionPane.showMessageDialog(this, "Null field or
     invalid number format", "Error", JOptionPane.ERROR_MESSAGE);
323                     }
324                 }
325             }
326         }
327
328         public class PrismWindow extends JFrame implements ActionListener
    {
329             private Container contentPane;
330             private JLabel lengthLabel, widthLabel, heightLabel,
    volumeLabel, surfaceLabel;
331             private JTextField lengthField, widthField, heightField;
332             private JButton calculateButton;
333
334             public PrismWindow() {
335                 start();
336                 setTitle("Rectangular Prism");
337                 setSize(280, 240);
338                 setLocationRelativeTo(null);
339                 setResizable(false);
340             }
341
342             private void start() {
```

```
                contentPane = getContentPane();
                contentPane.setLayout(null);

                lengthLabel = new JLabel("Length (cm):");
                lengthLabel.setBounds(20, 20, 135, 23);
                lengthField = new JTextField();
                lengthField.setBounds(120, 20, 135, 23);

                widthLabel = new JLabel("Width (cm):");
                widthLabel.setBounds(20, 50, 135, 23);
                widthField = new JTextField();
                widthField.setBounds(120, 50, 135, 23);

                heightLabel = new JLabel("Height (cm):");
                heightLabel.setBounds(20, 80, 135, 23);
                heightField = new JTextField();
                heightField.setBounds(120, 80, 135, 23);

                calculateButton = new JButton("Calculate");
                calculateButton.setBounds(120, 110, 135, 23);
                calculateButton.addActionListener(this);

                volumeLabel = new JLabel("Volume (cm³):");
                volumeLabel.setBounds(20, 140, 220, 23);

                surfaceLabel = new JLabel("Surface Area (cm²):");
                surfaceLabel.setBounds(20, 170, 220, 23);

                contentPane.add(lengthLabel);
                contentPane.add(lengthField);
                contentPane.add(widthLabel);
                contentPane.add(widthField);
                contentPane.add(heightLabel);
                contentPane.add(heightField);
                contentPane.add(calculateButton);
                contentPane.add(volumeLabel);
                contentPane.add(surfaceLabel);
            }

            @Override
            public void actionPerformed(ActionEvent event) {
                if (event.getSource() == calculateButton) {
                    try {
                        double l = Double.parseDouble(lengthField.getText
    ().trim());
                        double w = Double.parseDouble(widthField.getText()
    .trim());
                        double h = Double.parseDouble(heightField.getText
    ().trim());
                        Prism prism = new Prism(l, w, h);
                        volumeLabel.setText("Volume (cm³): " + String.
    format("%.2f", prism.getVolume()));
                        surfaceLabel.setText("Surface Area (cm²): " +
    String.format("%.2f", prism.getSurfaceArea()));
```

```
392                     } catch (Exception e) {
393                         JOptionPane.showMessageDialog(this, "Null field or
    invalid number format", "Error", JOptionPane.ERROR_MESSAGE);
394                     }
395                 }
396             }
397         }
398
399
```

Listing 5: Clases de Interfaz Gráfica para las Figuras